1. **Define the following terms?**
   i. Nested structures
   ii. Array of structures

   i. Nested Structures

   Definition:
   A nested structure in C is a structure that contains another structure as its member. This helps in logically grouping related data within a structure.

   Example:

```
struct Date {
    int day, month, year;
};

struct Student {
    int roll;
    char name[50];
    struct Date dob; // Nested structure
};
```

   ii. Array of Structures

   Definition:
   An **array of structures** is a collection of structure variables stored in a single array. It allows storing and managing data for **multiple entities** of the same type.

   Example:

```
struct Student {
    int roll;
    char name[50];
};

struct Student s[5]; // Array of 5 students
```

2. **Differentiate between Structure and Union.**

Point of Difference	**Structure (struct)**	**Union (union)**
Memory Allocation	Allocates separate memory for each member	Allocates shared memory for all members
Size	Size = sum of all members' sizes	Size = size of the largest member
Access	All members can be accessed at the same time	Only one member holds a valid value at a time
Usage	Used when all members need to	Used when only one member needs

	store values simultaneously	to store value at a time
Example Use Case	Student record (name, roll, marks, etc.)	Storing data of different types in the same memory (e.g., sensor readings)

3. **Describe the concept of Structure in C with example.**

   Definition:
   A **structure** in C is a user-defined data type that allows grouping of variables of **different data types** under a single name.
   It is used to represent a **record** — like a student, employee, or book — where data elements of different types (int, float, char, etc.) are related logically.

   Example:

```
#include <stdio.h>

struct Student {
    int roll;
    char name[50];
    float marks;
};

int main() {
    struct Student s1 = {101, "Anita", 92.5};

    printf("Roll: %d\n", s1.roll);
    printf("Name: %s\n", s1.name);
    printf("Marks: %.2f\n", s1.marks);

    return 0;
}
```

4. **Define a structure called cricket that will describe the following information: i) Player Name ii) Team Name iii) Batting Average Using cricket declare an array player with 40 Elements. Write a C program to read the information of all the 40 players and print the name of the player having the highest batting average.**

```
#include <stdio.h>
#include <string.h>

struct cricket {
    char playerName[50];
    char teamName[50];
    float battingAverage;
};

int main() {
```

```
    struct cricket player[40];
    int i, topIndex = 0;

    // Read information for 40 players
    for(i = 0; i < 40; i++) {
        printf("\nEnter details for Player %d\n", i+1);

        printf("Enter Player Name: ");
        scanf(" %[^\n]", player[i].playerName);  // read string with
spaces

        printf("Enter Team Name: ");
        scanf(" %[^\n]", player[i].teamName);

        printf("Enter Batting Average: ");
        scanf("%f", &player[i].battingAverage);
    }

    // Find player with highest batting average
    for(i = 1; i < 40; i++) {
        if(player[i].battingAverage >
player[topIndex].battingAverage) {
            topIndex = i;
        }
    }

    // Print the top player
    printf("\nPlayer with the Highest Batting Average:\n");
    printf("Name: %s\n", player[topIndex].playerName);
    printf("Team: %s\n", player[topIndex].teamName);
    printf("Average: %.2f\n", player[topIndex].battingAverage);

    return 0;
}
```

5. **Declare a structure having members of appropriate type Branch Name Roll No Marks in 8 subjects Write a C program to create an array of 30 structure variables and read all the members of each variable.**

```
#include <stdio.h>
#include <string.h>

struct Student {
    char name[50];
    char branch[30];
    int roll;
    float marks[8];
};

int main() {
    struct Student s[30];
    int i, j;
```

```
    // Input for 30 students
    for(i = 0; i < 30; i++) {
        printf("\nEnter details for Student %d:\n", i + 1);

        printf("Enter Name: ");
        scanf(" %[^\n]", s[i].name); // read string with spaces

        printf("Enter Branch: ");
        scanf(" %[^\n]", s[i].branch);

        printf("Enter Roll Number: ");
        scanf("%d", &s[i].roll);

        for(j = 0; j < 8; j++) {
            printf("Enter Marks for Subject %d: ", j + 1);
            scanf("%f", &s[i].marks[j]);
        }
    }

    // Optional: display entered data
    printf("\n--- Displaying Entered Student Data ---\n");
    for(i = 0; i < 30; i++) {
        printf("\nStudent %d:\n", i + 1);
        printf("Name: %s\n", s[i].name);
        printf("Branch: %s\n", s[i].branch);
        printf("Roll No: %d\n", s[i].roll);
        printf("Marks: ");
        for(j = 0; j < 8; j++) {
            printf("%.2f ", s[i].marks[j]);
        }
        printf("\n");
    }

    return 0;
}
```

6. **Suppose you have to create and maintain a Record Book of your company. Record Book consists of the following fields: (i) Employees' Name (ii) Id Number (iii) Salary Amount (iv) Designation. How do you create and maintain this Record Book for 150 Employees of your organization if your management asks to see the salary of any particular employee?**

```
#include <stdio.h>
#include <string.h>

#define SIZE 150

struct Employee {
    char name[50];
    int id;
    float salary;
    char designation[50];
};
```

```
int main() {
    struct Employee emp[SIZE];
    int i, searchId, found = 0;

    // Input details of 150 employees
    for(i = 0; i < SIZE; i++) {
        printf("\nEnter details for Employee %d:\n", i + 1);

        printf("Name: ");
        scanf(" %[^\n]", emp[i].name);

        printf("ID Number: ");
        scanf("%d", &emp[i].id);

        printf("Salary: ");
        scanf("%f", &emp[i].salary);

        printf("Designation: ");
        scanf(" %[^\n]", emp[i].designation);
    }

    // Search employee by ID
    printf("\nEnter the Employee ID to view salary: ");
    scanf("%d", &searchId);

    for(i = 0; i < SIZE; i++) {
        if(emp[i].id == searchId) {
            printf("\nEmployee Found!\n");
            printf("Name: %s\n", emp[i].name);
            printf("Designation: %s\n", emp[i].designation);
            printf("Salary: %.2f\n", emp[i].salary);
            found = 1;
            break;
        }
    }

    if(!found) {
        printf("Employee with ID %d not found.\n", searchId);
    }

    return 0;
}
```

7. **Create a structure to specify data on students give below: Roll Number, Name, Department, Course, and Year of admission. Assume that there are not more than 450 students in collage, Write a program to print names of all students who joined in a particular year.**

```
#include <stdio.h>

struct Student {
    int roll;
```

```
    char name[50];
    char department[50];
    char course[50];
    int yearOfAdmission;
};

int main() {
    struct Student students[450];
    int n, i, searchYear;

    printf("Enter number of students (max 450): ");
    scanf("%d", &n);

    // Input student details
    for(i = 0; i < n; i++) {
        printf("\nEnter details for student %d:\n", i + 1);

        printf("Roll Number: ");
        scanf("%d", &students[i].roll);

        printf("Name: ");
        scanf(" %[^\n]", students[i].name);

        printf("Department: ");
        scanf(" %[^\n]", students[i].department);

        printf("Course: ");
        scanf(" %[^\n]", students[i].course);

        printf("Year of Admission: ");
        scanf("%d", &students[i].yearOfAdmission);
    }

    // Ask for the year to search
    printf("\nEnter the year to list students who joined in that
year: ");
    scanf("%d", &searchYear);

    printf("\nStudents admitted in %d:\n", searchYear);
    int found = 0;
    for(i = 0; i < n; i++) {
        if(students[i].yearOfAdmission == searchYear) {
            printf("%s\n", students[i].name);
            found = 1;
        }
    }

    if(!found) {
        printf("No students found for the year %d.\n", searchYear);
    }

    return 0;
}
```